

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and deployed **ML/AI models** for predictive maintenance and anomaly detection, reducing unscheduled downtime by **18%** across 500+ mining units.
- Architected a **standardized data baseline** for 10TB of daily telemetry from mining sites, structuring raw data for AI/ML processing and ensuring **99.9% data quality**.
- Developed and integrated **optimization algorithms** into the Mining Operating System (MOS), enhancing energy efficiency and improving overall site performance by **12%**.
- Led the definition and execution of the **AI strategy** for MOS and MDK platforms, translating operational challenges into solvable data and optimization problems.
- Engineered **Python automation scripts** for data pipeline management and ETL processes, reducing manual data preparation time by **40%** and enabling faster model iteration.
- Collaborated with backend engineers to evolve data pipeline and integration standards, embedding **AI models into platform workflows and APIs** with **95%** uptime.

SKILLS

AI/ML & Optimization – Machine Learning, Predictive Maintenance, Anomaly Detection, Performance Optimization, Mathematical Optimization, Data Modelling, Statistical Analysis, Regression, Time-Series Analysis
Data Engineering & Architecture – Data Baseline Design, Data Schema Design, Data Warehousing, ETL, Data Pipelines, SQL (PostgreSQL, MySQL, Oracle), Data Quality Assurance
Programming & Scripting – Python, Java, Spring Boot, REST APIs, R
Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux
Development Tools & Methodologies – Git, Jenkins, Jira, Confluence, Agile/Scrum, CI/CD
Business Intelligence – Power BI, Tableau, DAX, Power Query
Domain Specific – Bitcoin Mining Operations, Industrial IoT Telemetry, Operational Efficiency, Energy Efficiency

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Mining Operating System (MOS) AI-Driven Optimization Platform

- Architected and implemented a comprehensive **AI-driven optimization platform** for a global Bitcoin mining network, processing over **10TB of daily telemetry data**.

- Developed **predictive maintenance models** using Python and scikit-learn, identifying potential hardware failures **72 hours in advance** and reducing critical incidents by **25%**.
- Designed and deployed **optimization algorithms** leveraging mathematical modelling to enhance energy efficiency, resulting in a sustained **15% reduction in operational costs** across multiple sites.

Financial Services Transaction Monitoring Dashboard

- Developed and maintained a **payments operations dashboard** for a financial services client, providing real-time transaction-flow monitoring and operational alerting.
- Implemented **automated reconciliation workflows** using Python and SQL, improving the accuracy of financial data matching by **99%** and reducing manual reconciliation effort.
- Ensured production environments adhered to strict SLAs and audit trail requirements, supporting **GDPR-aware data handling** and operational continuity.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)